

LAB REPORT
FOR
LAB – 3: SYNCHRONIZATION
EEC 556: SOFTWARE DEFINED RADIO
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SYNCHRONIZATION

SYMBOL TIMING RECOVERY IN NARROWBAND CHANNELS

INTRODUCTION:

In this lab, we are looking at one of the main synchronization problems in a communication system, that of symbol synchronization. The received signal undergoes a number of distortions and delays as it is received by the receiver – in addition to being corrupted by noise.

The received signal can be expressed by $z(t) = \alpha e^{j\phi} x(t - \tau d) + v(t)$

where α is the attenuation, ϕ is the phase shift, $v(t)$ is the AWGN, and d is the delay.

we have to estimate τd so that we can shift the received signal by the same amount to essentially get rid of this delay. If this delay is greater than the symbol duration, we need synchronization at a higher level, that of frame synchronization.

This delay in discrete time by determining an amount of delay $\hat{k}$ and then to delay the filtered received signal by $\hat{k}$ prior to down sampling. Two methods will be used for symbol synchronization in this lab: the maximum energy method and the Early Late gate method.

BLOCK DIAGRAMS:

- ❖ We have downloaded lab3 files and started working on them, first we have modified max energy with using lab manual and given lab notes.
- ❖ Max energy block diagram after modifying with our version and saving it with new names can be shown in figure 1

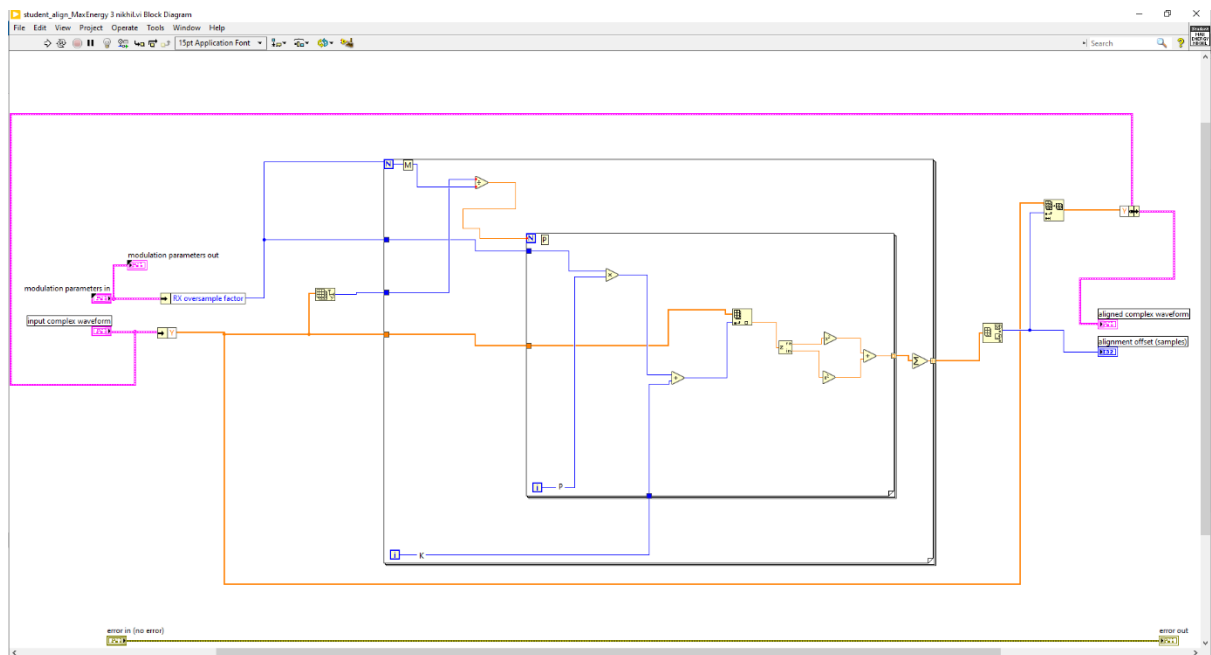


Figure 1: student_align_maxEnergy nikhil.vi(Block diagram)

- ❖ After finishing max energy we worked on El gate and modified it with using lab manual and lab notes.
- ❖ El gate block diagram after modifying with our version and saving with new names can be shown in figure 2

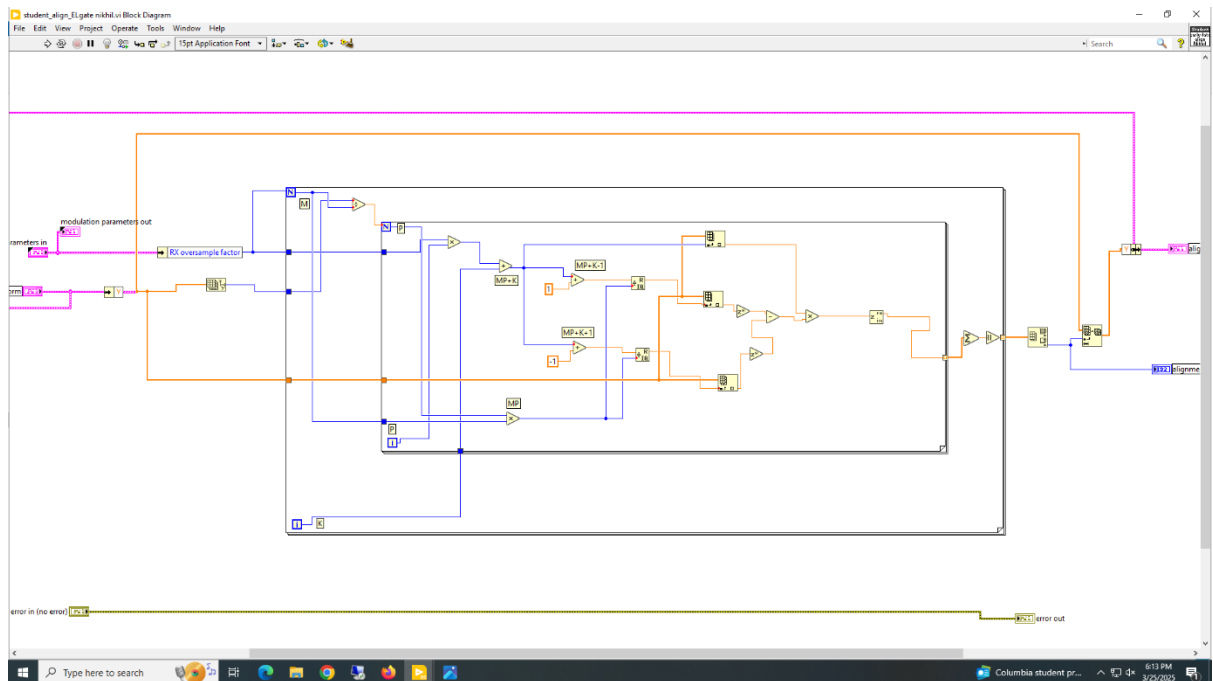


Figure 2: student_align_Elgate nikhil.vi(Block diagram)

- ❖ After modifying both max energy and El gate we have replaced both vi's as shown in lab notes from receiver.vi double click simulator.vi in that select time estimation case open symbol_timing.vi from that selecting maximum energy case and replaced with student_align_max_energy nikhil.vi and after selecting early late gate case replaced with student_align_elgate nikhil.vi as shown in figure 3 and figure 4

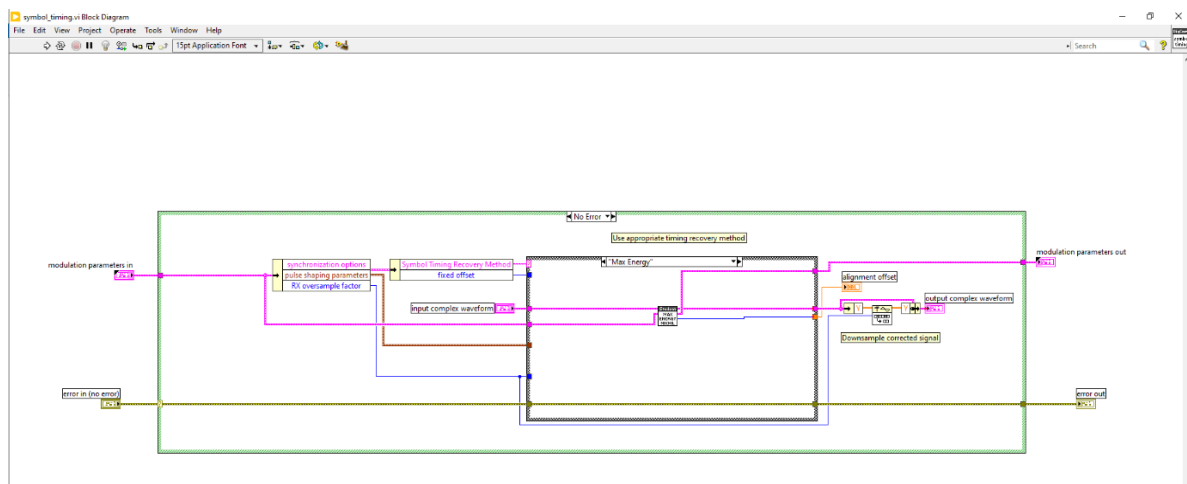


Figure 3: symbol_timing.vi(Replaced max energy)

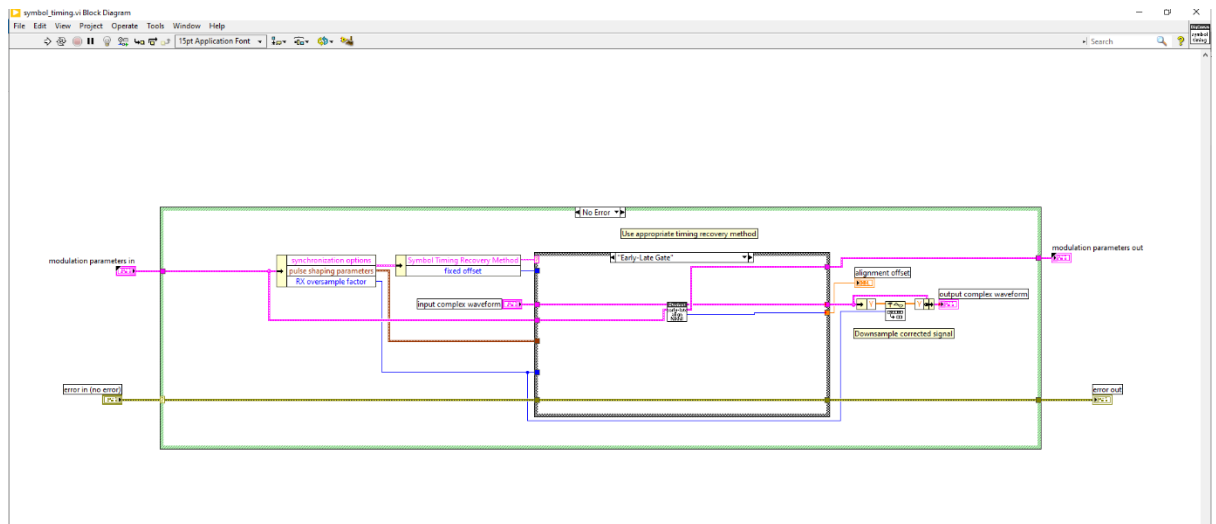


Figure 4: symbol_timing.vi(Replaced El gate)

- ❖ After replacing both Max energy and El gate we have observed simulation results in simulator.vi as shown in figure 5,6,7,8,9,10.

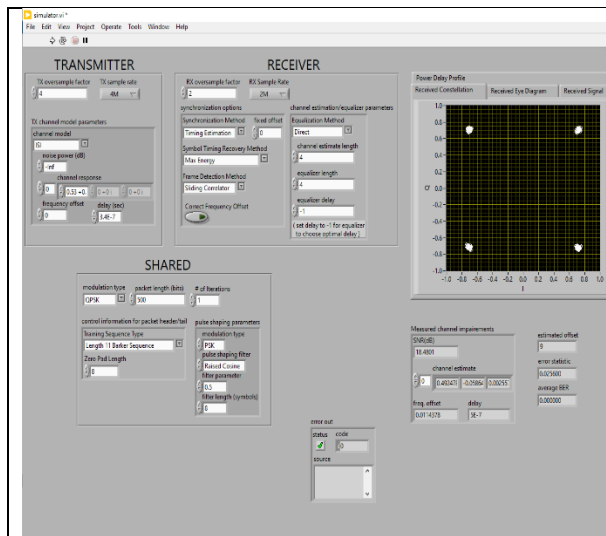


Figure 5: simulator.vi(max energy 1)

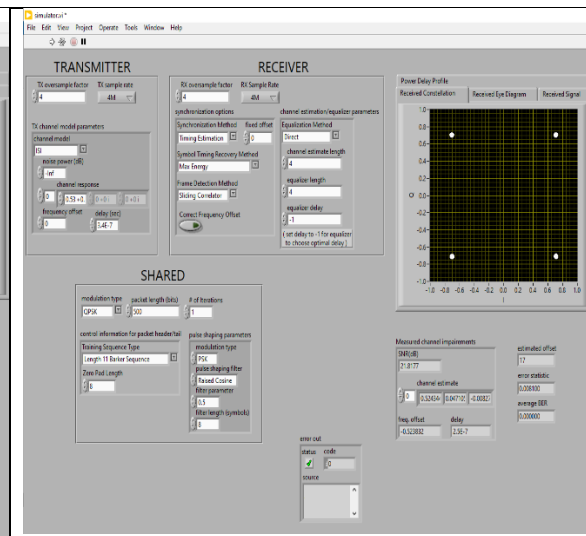


Figure 6: simulator.vi(max energy 2)

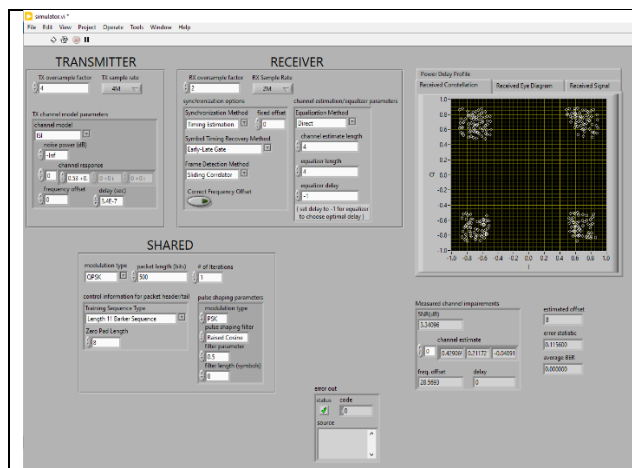


Figure 7: simulator.vi(El gate 1)

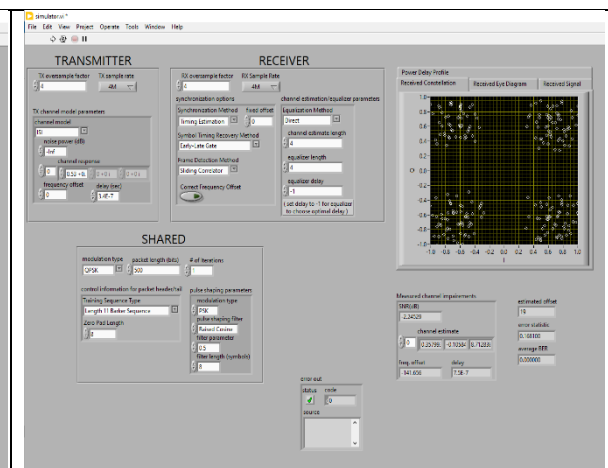


Figure 8: simulator.vi(El gate 2)

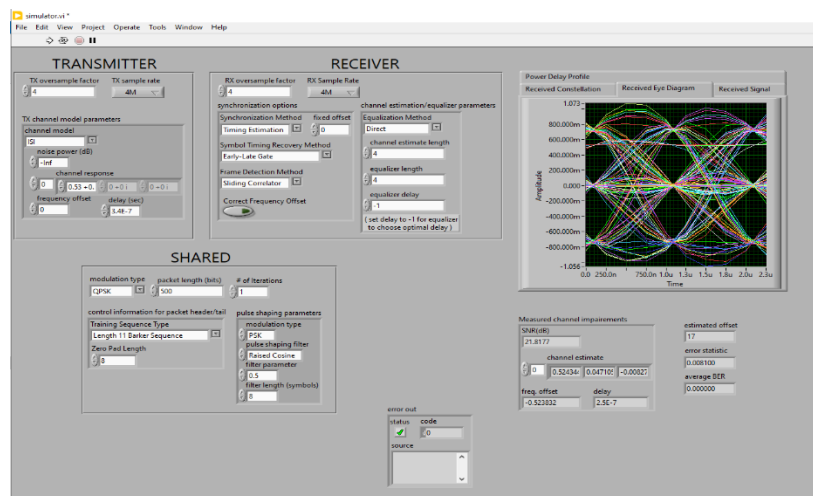


Figure 9: simulator.vi(Received Eye diagram)



Figure 10: simulator.vi(Received signal)

We have set following parameters on simulator.vi to observe plot between Timing error statistic Vs Oversampling factor:

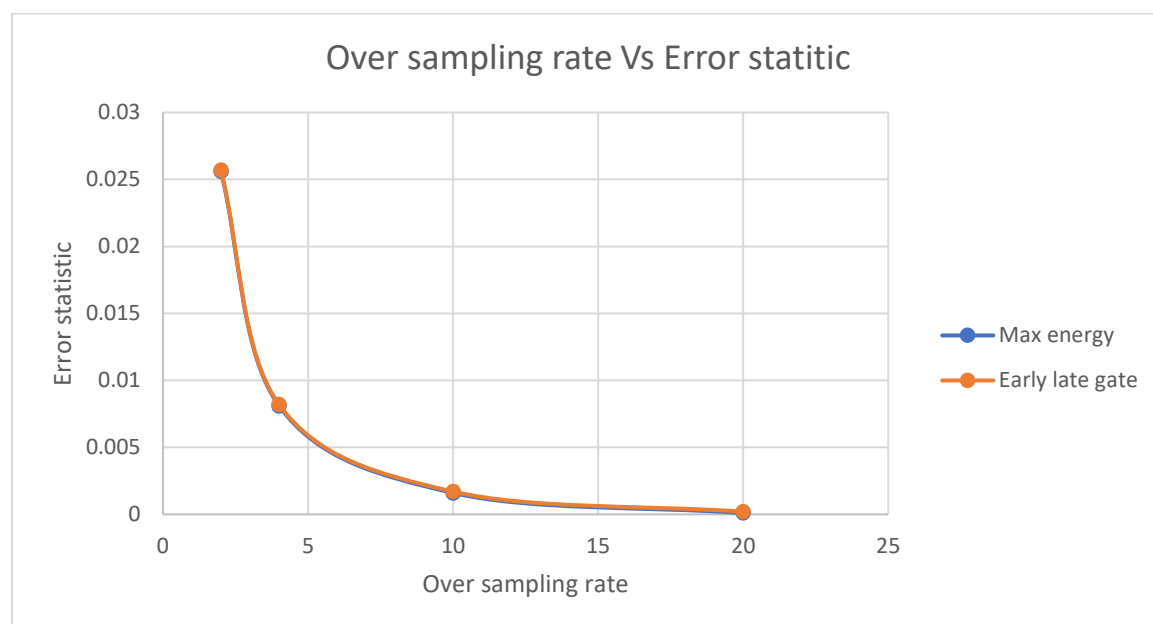
$$\text{Delay } \tau_d = 3.4E - 7$$

Oversample Factor N = 2, 4, 6, 10, 20

Channel Model: $h = 0.53+0.53i$

Noise Power: $-\infty$ dB

Over sampling rate	2	4	10	20
Error statistic -Max energy	0.0256	0.0081	0.0016	0.0001
Error statistic-Early late gate	0.0256	0.0081	0.0016	0.0001



LAB EXPERIMENT:

For the lab experiment after connecting NI-Usrcp devices to computer we run top_tx.vi and top_rx.vi after setting following parameters

Packet Length = 500 bits

Modulation type = QPSK

Pulse shaping filter = Root Raised

Cosine Filter Parameter = 0.5

Filter length = 8

Channel model $h = 0.75e^{j\pi/4}$ which is $(0.53 + j0.53)$

Noise Power = -Inf dB

Tx sample rate = 20 MSamples/s

Tx Oversampling filter = 20

Rx sample rate = 2 MSamples/s

Rx Oversampling filter = 2Symbol

Timing recovery method = Max Energy

Capture time = 400 μ sec

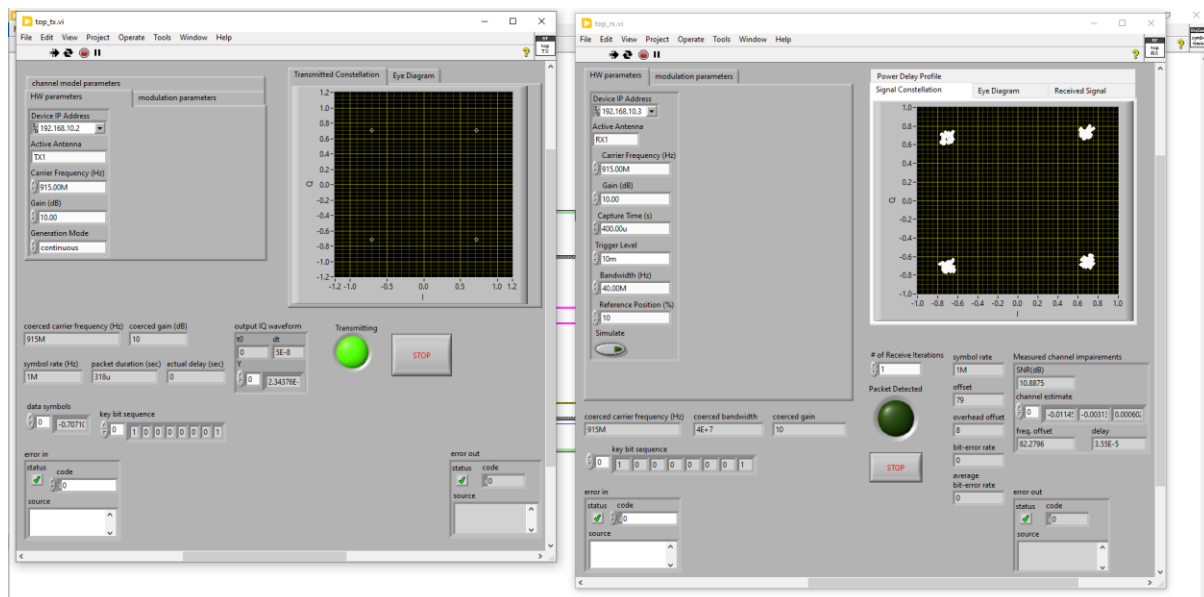


Figure 11: top_tx.vi & top_rx.vi (transmitted constellation and signal constellation)

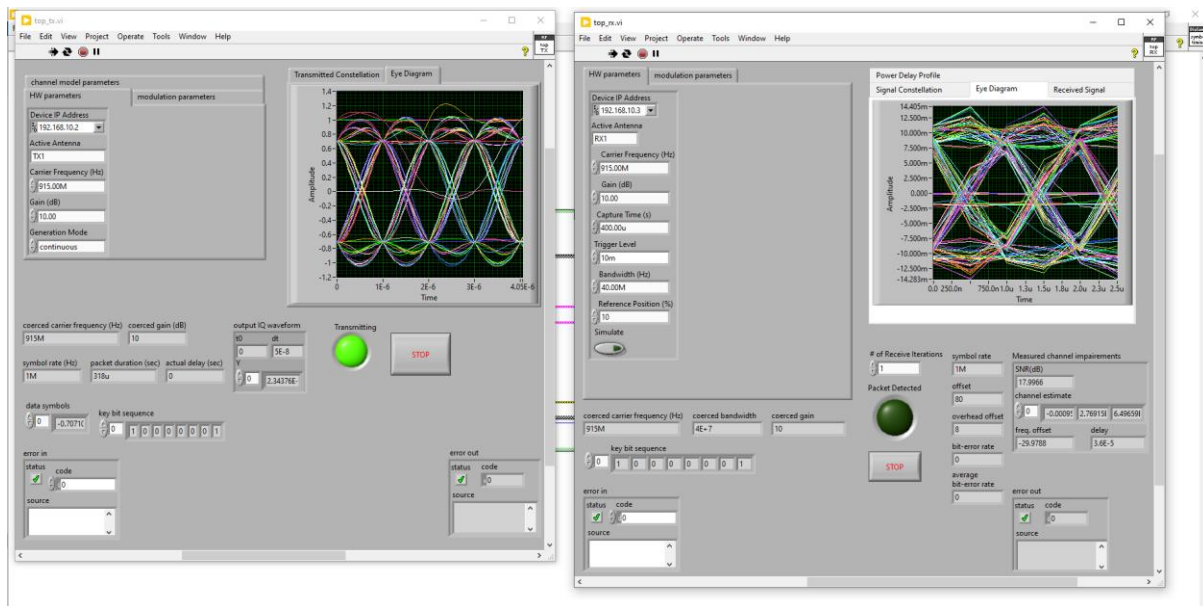


Figure 12: top_tx.vi & top_rx.vi (Eye diagram)

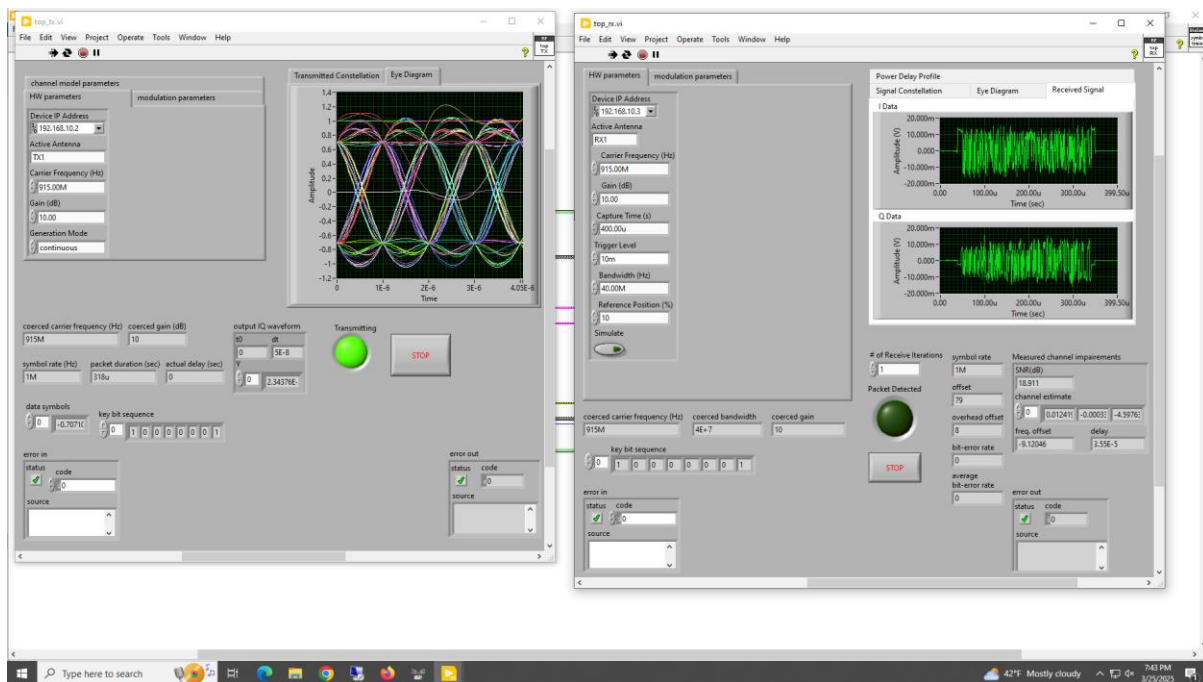


Figure 13: top_tx.vi & top_rx.vi(Received signal)

QUESTION AND ANSWERS:

- **What are the two critical assumptions used to formulate the indirect maximization of the output energy? Consider how the presence of the flat fading channel AWGN can impact this method. Specifically, using at least one of the critical assumptions, explain how you might mitigate the impact of these impairments by suitable selection of parameters.**

Ans: There are several assumptions made to formulate indirect maximization of the output as follows:

- The pulse shape is assumed to be Nyquist pulse shape.
 - The value of τ_d is assumed to be fraction of symbol period thus $0 < \tau_d < T$
 - The discrete time
 - This approach is consider based on the assumption of it can be differentiate and discretizing the result is possible on later stage.
- **After down sampling a sequence originally sampled at rate $1/T_z$ by a factor M , what is the sample period of the resulting signal?**

Ans: T_z / M (reciprocal of symbol rate and same as transmitted rate for ideal channel)
Symbol rate (so as symbol period should be same at transmitter and receiver in ideal channel)

As Symbol rate is reciprocal of Symbol period, if the sequence is originally sampled at rate $1/T_z * M$.

- **What is the symbol rate of the system based on the parameters above?**
Ans: $1M$ samples/second (as Symbol rate = sample rate / Over-sampling factor, here $2M/2$, or $4M/4 \dots 20M/20$).
- **As you did in the pre-lab, vary the oversample factor N at the receiver. Observe the general shape of the received constellation for a number of packets (i.e., to eliminate any variance that might be caused by errors in frequency offset correction or channel estimation). Observe how the constellation changes when $N=2, 4, 10$ and 20 .**

Based on what you've observed and what you learned in the pre-lab, describe how the relationship between sampling error and oversample factor manifests itself on the signal constellation of `top_rx.vi`.

Ans: The relationship between sampling error and oversample factor becomes evident when running the program. For $N=2$, the received constellation appears less accurate and is somewhat dispersed. However, as we increase N to values like 4, 10, and 20, the results become more precise, and the received constellation presents a clearer view. This indicates that increasing the N value reduces sampling error, leading to more accurate results.

- **For each of the oversample factors in the previous question (i.e., $N=2, 4, 10$ and 20), specify what value you set the sample rate to on the front panel of `top_rx.vi`.**

Ans: 2, 4, 10, 20 (M Sample/Second) for over-sampling values of $N=2, 4, 10, 20$ respectively.

- **Discuss any problems you might have encountered and how you overcame these obstacles.**

Ans: while replacing both Max energy and El gate we have encountered problem , we have overcame the problem with carefully reading given lab notes.

- **What underlying assumption about the noise in the system has been made, but not stated explicitly in this lab? Explain why this assumption is important for the maximum output energy maximization problem.**

Ans: The system is modeled as Additive White Gaussian Noise (AWGN), where noise is added to the signal. If the system were not AWGN, distinguishing between the signal and noise would complicate the energy maximization process. The presence of AWGN facilitates optimal filter design, making it easier to separate the signal from the noise.

- **Air any grievances about the structure, content, or workload for this lab.**

Ans: In my view, this lab was definitely more challenging than the previous two. It really pushed me to tap into my background knowledge and logical reasoning before diving into the programming part. The implementation process was particularly tricky, as it required specific tools and icons to create a design without errors.

Moreover, the indirect method we used to maximize energy output added an extra layer of complexity, making the whole execution process even more difficult. Overall, it was a tough but valuable experience.